

Safe Adaptive Feedback Control via Barrier States[★]

Trivikram Satharasi^{*} Tochukwu E. Ogri^{*} Muzaffar Qureshi^{*}
 Kyle Volle^{**} Rushikesh Kamalapurkar^{*}

^{*} *University of Florida, Mechanical and Aerospace Engineering,
 Gainesville, FL 32608 USA (e-mail: t.satharasi@ufl.edu,
 tochukwu.ogri@ufl.edu, muzaffar.qureshi@ufl.edu, rkamalapurkar@ufl.edu.)*
^{**} *Torch Technologies, Shalimar, FL, USA (email:
 Kyle.Volle@torchtechnologies.com.)*

Abstract: This paper presents a safe feedback control framework for nonlinear control-affine systems with parametric uncertainty by leveraging adaptive dynamic programming (ADP) with barrier-state augmentation. The developed ADP-based controller enforces control invariance by optimizing a value function that explicitly penalizes the barrier state, thereby embedding safety directly into the Bellman structure. The near-optimal control policy computed using model-based reinforcement learning is combined with a concurrent learning estimator to identify the unknown parameters and guarantee uniform convergence without requiring persistency of excitation. Using a barrier-state Lyapunov function, we establish boundedness of the barrier dynamics and prove closed-loop stability and safety. Numerical simulations on an optimal obstacle-avoidance problem validate the effectiveness of the developed approach.

Keywords: Safe optimal control; Adaptive dynamic programming; Barrier states.

1. INTRODUCTION

Control barrier functions (CBFs) have become a popular tool for enforcing safety, see Ames et al. (2014). While initial work on CBFs was model-based, recent research has focused on extending CBF-based feedback control to ensure safety under model uncertainty (Mahmud et al. (2021); Ogri et al. (2025)).

To guarantee safety and stability using reinforcement learning (RL), Cohen and Belta (2023) propose a decoupled safety-aware controller by developing a dedicated safety controller which is paired with a model-based reinforcement learning (MBRL) controller, allowing the system to maintain safety without compromising stability (Kamalapurkar et al. (2016a)). However, the controllers in the work are performance-driven but not optimal. Furthermore, RL-Quadratic Programming (QP) based controllers, such as in Cohen et al. (2023); Krstic (2023), ensure safety by solving a quadratic program at each step, but this safety enforcement can modify the control input in ways that deviate from the original MBRL policy, potentially compromising the stability properties that the MBRL controller would otherwise guarantee. Recent developments have further shown that CBF-based quadratic programs can introduce multiple undesirable asymptotically stable equilibria (Reis et al. (2021)).

Another popular approach to safe control is the barrier transformation (BT) method (Greene et al. (2020); Mahmud et al. (2021)), which applies nonlinear coordinate transformations to system states. Unlike CBF-based techniques, the BT approach reformulates the constrained state-feedback control problem into an equivalent unconstrained problem through a state transformation, enabling the direct design of controllers that inherently satisfy state constraints. Although BT-based approaches yield verifiable safe state-feedback controllers, they are limited to box constraints.

To address these limitations, Almubarak et al. (2025) developed a barrier operator that converts safety constraints into an intrinsic coordinate – the barrier state (BaS). Bounded BaS guarantees a forward-invariant safety cone in the augmented dynamics. Compared with CBFs, no explicit CBF synthesis is needed; unlike BT-based approaches, the method handles a wider class of constraints. Both BaS and CBF methods require full knowledge of the dynamics, which is rarely available in practice. Robust CBF extensions for uncertain systems (Taylor and Ames (2020); Lopez et al. (2020)) often impose conservative control bounds. Embedding BaS with adaptive parameter estimation (Aoun et al. (2023); Al-Sunni et al. (2025)) preserves safety under uncertainty; nevertheless, works such as Al-Sunni et al. (2025) do not ensure the estimates converge to their true values, leaving a gap between robust safety and accurate identification.

The main contributions of this work are as follows. First, by embedding the safety constraint into an augmented system state, the BaS-RL framework ensures that the controller maintains near-optimal safety even when the system

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parameters are unknown. Second, by integrating an actor-critic ADP framework along with an Integral Concurrent Learning (ICL) based parameter estimator, our approach learns the optimal value function and generates a control policy that balances safety, stability, and performance. Through a Lyapunov-based analysis, we show that the developed ICL-BaS-RL architecture simultaneously achieves both safety, and the estimation of parameters. Because the BaS state is virtually propagated for control design, unknown system dynamics prevent direct integration of BaS, ADP, and ICL. We address this by introducing an observer that uses parameterized dynamics estimates and corrects the estimated barrier state with the measured barrier state.

2. PROBLEM FORMULATION

Consider a nonlinear control-affine system of the form

$$\dot{x} = Y(x)\theta + f(x) + g(x)u, \quad x(0) = x_0, \quad (1)$$

where $x \in \mathbb{R}^n$ is system state vector, $u \in \mathbb{R}^m$ is the control input, $\theta \in \mathbb{R}^p$ is a vector of unknown parameters, and $Y : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times p}$, $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $g : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are the known system regressor matrix, drift dynamics, and control effectiveness matrix respectively.

Assumption 1. The functions f , g , and Y are locally Lipschitz continuous and satisfy $f(0) = 0$, $Y(0) = 0$, and $0 < \|g(x)\| \leq \bar{G}$ for some $\bar{G} > 0$ and for all $x \in \mathbb{R}^n$.

Assumption 2. There exists a known $\bar{\theta} > 0$ s.t. $\|\theta\| \leq \bar{\theta}$.

Let $\mathcal{S} \subset \mathbb{R}^n$ represent the zero super-level set of a continuously differentiable function $h \in \mathcal{C}^1(\mathbb{R}^n; \mathbb{R})$, defined as

$$\mathcal{S} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\}, \quad (2)$$

$$\partial\mathcal{S} = \{x \in \mathbb{R}^n \mid h(x) = 0\}, \quad (3)$$

$$\text{int}(\mathcal{S}) = \{x \in \mathbb{R}^n \mid h(x) > 0\}, \quad (4)$$

where $\partial\mathcal{S}$ and $\text{int}(\mathcal{S})$ denote the boundary and interior of $\mathcal{S}$, respectively.

Assumption 3. The set $\mathcal{S}$ is nonempty, contains no isolated points, and $\nabla h(x) \neq 0, \forall x \in \partial\mathcal{S}$.

Let $t \mapsto x(t)$ denote the trajectory of (1) starting from x_0 under the input $u(\cdot)$.

Definition 1. A set $\mathcal{S} \subset \mathbb{R}^n$ is said to be conditionally control invariant with respect to $\mathcal{S}_0 \subseteq \mathcal{S}$ if $\exists u(\cdot) \in \mathcal{U}$ s.t., $\forall x_0 \in \mathcal{S}_0$ and $\forall t \geq 0$, $x(t) \in \mathcal{S}$.

The objective of this paper is to estimate the unknown parameters of the system (1) while simultaneously synthesizing a safe feedback control policy that ensures conditional control invariance as described in Definition 1.

3. SAFETY-EMBEDDED MODEL

To integrate constraint-awareness directly into the system model in a manner that preserves differentiability, supports observer and estimator design, and admits Lyapunov-based guarantees, consider a barrier operator $B \in \mathcal{C}^\infty((0, \infty); \mathbb{R})$ such that $\frac{dB(a)}{da} < 0 \quad \forall a > 0$, $\lim_{a \rightarrow 0^+} B(a) = \infty$, and $\frac{dB(a)}{da} \circ B^{-1} \in \mathcal{C}^\infty((0, \infty); \mathbb{R})$. For the continuously differentiable function $h \in \mathcal{C}^1(\mathbb{R}^n; \mathbb{R})$ that defines $\mathcal{S}$, define the composed barrier function

$$\beta(x) := B(h(x)). \quad (5)$$

Because B diverges as $h(x) \rightarrow 0^+$, β satisfies $\beta(x) \geq 0, \forall x \in \text{int}(\mathcal{S})$ and $\lim_{x \rightarrow \partial\mathcal{S}} \beta(x) = \infty$, so the safety condition $h(x) > 0$ becomes equivalent to $\beta(x) < \infty$.

Using a coordinate transformation, the BaS is defined as $z := \beta(x) - \beta_0$ with $\beta_0 := \beta(0)$. The dynamics in these coordinates become

$$\dot{z} = \Phi(z + \beta_0) \nabla h(x) (Y(x)\theta + f(x) + g(x)u). \quad (6)$$

where $\Phi = \frac{B(h)}{dh} \circ B^{-1} \in \mathcal{C}^\infty((0, \infty); \mathbb{R})$ appears as a nonlinear gain induced by the barrier operator, and the initial BaS is $z_0 = \beta(x_0) - \beta_0$.

The BaS induces a forward-invariant cone in the extended (x, z) -space that encodes safety as boundedness of z .

Lemma 1. Let $\mathcal{C} := \{(x, z) \in \mathbb{R}^n \times \mathbb{R} \mid z = \beta(x) - \beta_0\}$. Consider any complete solution $(x(\cdot), z(\cdot))$ of (1) and (6) under a feedback $(t, s) \mapsto u(t, s)$ that is locally Lipschitz continuous in x , and piecewise continuous in t . Then $z(t) = \beta(x(t)) - \beta_0$ for all $t \geq 0$ if and only if $(x(0), z(0)) \in \mathcal{C}$. Moreover, for any $t \geq 0$, $x(t) \in \mathcal{S}$ if and only if $\beta(x(t)) < \infty$ i.e. $z(t) < \infty$.

Proof. Differentiate $t \mapsto \beta(x(t)) - \beta_0$ along solutions and compare with $\dot{z}$ in (6). By uniqueness of solutions, the two scalar trajectories coincide whenever they agree at $t = 0$. The final equivalences follow from (Almubarak et al., 2025, Theorem 1) and the defining property $\beta(x) = \infty$ if and only if $x \notin \mathcal{S}$. $\square$

Lemma 1 formalizes that the safety condition is dynamically equivalent to forward-boundedness of a lifted coordinate, allowing the use of Lyapunov arguments directly on z rather than on $\beta(x)$.

Since θ in (6) is unknown, we introduce the estimated BaS, denoted by $\hat{z} \in \mathbb{R}$, which evolves according to

$$\dot{\hat{z}} = \Phi(z + \beta_0) \nabla h(x) (Y(x)\hat{\theta} + f(x) + g(x)u) + \gamma(z - \hat{z}), \quad (7)$$

where $\gamma > 0$ is a user-defined observer gain. Let the BaS estimation error be defined as $\tilde{z} := z - \hat{z}$. Subtracting (7) from (6) yields the error dynamics

$$\dot{\tilde{z}} = \Phi(z + \beta_0) \nabla h(x) Y(x)\tilde{\theta} - \gamma\tilde{z}, \quad (8)$$

where $\tilde{\theta} := \theta - \hat{\theta}$ is the parameter estimation error. The next section introduces a parameter-estimation law tailored to (8) that preserves the forward-invariant cone of Lemma 1.

4. PARAMETER ESTIMATOR DESIGN

In this section, an ICL update law (Chowdhary and Johnson (2010)) is developed to estimate the unknown parameters. The parameter estimator relies on the fact that the difference between the estimated and the true BaS at time t and time $t - T$ can be expressed as an affine function of the parameters θ and a residual that reduces with reducing BaS estimation error.

Lemma 2. For any fixed delay $T > 0$ and all $t \geq T$, the system dynamics satisfy the incremental relation

$$\mathcal{X}(t) = \mathcal{Y}(t)\theta + \mathcal{G}_{fu}(t), \quad (9)$$

where $\mathcal{X}(t) := x(t) - x(t - T)$, $\mathcal{Y}(t) := \int_{t-T}^t Y(s(\tau)) d\tau$, and $\mathcal{G}_{fu}(t) := \int_{t-T}^t (f(s(\tau)) + g(s(\tau))u(\tau)) d\tau$.

Proof. The result follows directly from the Fundamental Theorem of Calculus applied to (1). $\square$

Lemma 2 implies that, for any time instant τ , the parametric regression error satisfies

$$\mathcal{Y}(t)\tilde{\theta}(\tau) = \mathcal{X}(t) - \mathcal{G}_{fu}(t) - \mathcal{Y}(t)\hat{\theta}(\tau), \quad (10)$$

which motivates the following ICL update law

$$\dot{\hat{\theta}} = \text{proj}_{\Theta}(\hat{\theta}, \phi, \Gamma), \quad (11)$$

where $\Gamma \in \mathbb{R}^{p \times p}$ is a time-varying positive definite least-squares gain matrix that satisfies

$$\dot{\Gamma} = \begin{cases} \beta_{\theta}\Gamma - k_{\theta}\Gamma \sum_{i=1}^N \frac{\mathcal{Y}_i^{\top}\mathcal{Y}_i}{1+\kappa\|\mathcal{Y}_i\|^2}\Gamma, & \text{if } \hat{\theta} \in \text{int}(\Theta) \\ & \text{or } (\hat{\theta} \in \partial\Theta \text{ and } \phi^{\top}\hat{\theta} \leq 0), \\ 0 & \text{otherwise,} \end{cases}$$

and $\phi := \Gamma k_{\theta} \sum_{i=1}^N \frac{\mathcal{Y}_i^{\top}(\mathcal{X}_i - \mathcal{G}_{fu,i} - \mathcal{Y}_i\hat{\theta})}{1+\kappa\|\mathcal{Y}_i\|^2} + \Gamma Y(x)^{\top} \nabla h(x)^{\top} \Phi(z + \beta_0)^{\top} \tilde{z}$, with $\mathcal{X}_i := \mathcal{X}(t_i)$, $\mathcal{Y}_i := \mathcal{Y}(t_i)$, $\mathcal{G}_{fu,i} := \mathcal{G}_{fu}(t_i)$. The gains $k_{\theta}, \beta_{\theta}, \kappa \in \mathbb{R}_{>0}$ are user-defined adaptation gains, and $\text{proj}_{\Theta}(\cdot, \cdot)$ is the smooth projection operator introduced in Appendix E of Krstic et al. (1995). Given $\lambda_{\min}\{\Gamma^{-1}(0)\} > 0$, by using arguments similar to those presented in (Ioannou and Sun, 1996, Section 4.4.2), the update law in (4) ensures that the least-squares update law satisfies $\underline{\Gamma}I_p \leq \Gamma(t) \leq \bar{\Gamma}I_p$, $\forall t \in \mathbb{R}_{\geq 0}$ for some $\underline{\Gamma}, \bar{\Gamma} > 0$, where I_p denotes a $p \times p$ identity matrix. To facilitate data reuse, a finite set of delayed samples is maintained in a *history stack* $\mathcal{H} := \{\mathcal{X}_i, \mathcal{Y}_i, \mathcal{G}_{fu,i}\}_{i=1}^N$, which is updated at discrete times t_i based on data persistence conditions. The corresponding dynamics follow from (11) as

$$\dot{\hat{\theta}} = -\text{proj}_{\Theta}(\hat{\theta}, (k_{\theta}\Gamma\Sigma_{\mathcal{Y}}\hat{\theta} + \Gamma Y(x)^{\top} \nabla h(x)^{\top} \Phi(z + \beta_0)^{\top} \tilde{z})),$$

where $\Sigma_{\mathcal{Y}} \in \mathbb{R}^{p \times p}$ denotes the weighted regressor matrix defined as $\Sigma_{\mathcal{Y}} := \sum_{i=1}^N \sigma_i \mathcal{Y}_i^{\top} \mathcal{Y}_i$ with $\sigma_i = \frac{1}{1+\kappa\|\mathcal{Y}_i\|^2} > 0$.

Assumption 4. There exists a finite collection of sampling instants $\{t_i\}_{i=1}^N$ such that for all $t > 0$,

$$\lambda_{\min}(\Sigma_{\mathcal{Y}}) =: \underline{\sigma}_{\theta} > 0. \quad (12)$$

Remark 1. Assumption 4 is commonly adopted in ICL-based parameter estimation frameworks (Chowdhary and Johnson (2010); Parikh et al. (2019); Ogri et al. (2025)) and entails that the system undergo sufficient excitation. Unlike the classical PE condition, however, it only demands excitation over a finite interval, rendering it a weaker requirement.

A history stack satisfying (12) is said to be *full rank*. The constant $\underline{\sigma}_{\theta}$ quantifies the informational richness of the stored data and directly influences the rate of convergence of (11). To guarantee sufficient excitation and improve estimator conditioning, the history stack is updated using a minimum-eigenvalue maximization strategy (Chowdhary and Johnson (2010)). A candidate data point $(\mathcal{X}^*, \mathcal{Y}^*, \mathcal{G}_{fu}^*)$ is admitted by replacing an existing entry $(\mathcal{X}_j, \mathcal{Y}_j, \mathcal{G}_{fu,j})$ if it strictly increases the smallest eigenvalue of $\Sigma_{\mathcal{Y}}$, i.e.,

$$\lambda_{\min}(\Sigma_{\mathcal{Y}}^o) < \frac{\lambda_{\min}(\Sigma_{\mathcal{Y}}^*)}{1+\delta}, \quad \delta > 0, \quad (13)$$

where $\Sigma_{\mathcal{Y}}^o = \sum_{i \neq j} \sigma_i \mathcal{Y}_i^{\top} \mathcal{Y}_i + \sigma_j \mathcal{Y}_j^{\top} \mathcal{Y}_j$, $\Sigma_{\mathcal{Y}}^* = \sum_{i \neq j} \sigma_i \mathcal{Y}_i^{\top} \mathcal{Y}_i + \sigma^* \mathcal{Y}^{*\top} \mathcal{Y}^*$, and $\sigma^* = \frac{1}{1+\kappa\|\mathcal{Y}^*\|^2}$. By construction, (13) ensures that the minimum eigenvalue of $\Sigma_{\mathcal{Y}}$ is non-decreasing

over time. In the following section, we develop a stabilizing control law for the augmented system in (14) that minimizes the cost functional introduced in (15) and satisfies the safety requirements of the original system (1).

5. CONTROL DESIGN

(Almubarak et al., 2025, Theorem 1) and Lemma 1 indicate that the boundedness of BaS implies safety. Hence to design a control policy that enforces safety, we enforce boundedness of the BaS by augmenting it to the state space of the system. The corresponding augmented state is denoted by $s = [x^{\top}, z]^{\top}$ with the dynamics taking the form

$$\dot{s} = A(s)\theta + F(s) + G(s)u, \quad s(0) = s_0, \quad (14)$$

where $A(s) := [Y(x)^{\top}, \Phi(z + \beta_0)^{\top} \nabla h(x)^{\top} Y(x)^{\top}]^{\top}$, $F(s) = [f(x)^{\top}, \Phi(z + \beta_0)^{\top} \nabla h(x)^{\top} f(x)^{\top}]^{\top}$, and $G(s) = [g(x)^{\top}, \Phi(z + \beta_0)^{\top} \nabla h(x)^{\top} g(x)^{\top}]^{\top}$. The construction of z ensures that conditional control invariance of the safety set $\mathcal{S}$ is equivalent to local uniform boundedness of the $(n+1)^{\text{st}}$ coordinate of s .

5.1 Optimal Control Problem

Let $Q \in \mathbb{S}_+^{n+1}$ and $R \in \mathbb{S}_+^m$, where $\mathbb{S}_+^n$ is the space of positive definite square matrices of order n . For a given control input $u : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^m$, define the cost functional

$$J(s_0, u) = \int_0^{\infty} s(\tau)^{\top} Q s(\tau) + u(\tau)^{\top} R u(\tau) d\tau, \quad (15)$$

where $s(\cdot)$ is the solution of (14) starting from initial state s_0 and under the control input $u(\cdot)$.

Remark 2. In the following, we assume that a solution of the optimal control problem exists and the system (14) is stabilizable. If the system is not stabilizable, then the dynamic compensation technique proposed in Almubarak et al. (2025) can be employed to ensure stabilizability.

The value function of the optimal control problem to minimize J is defined as $V^*(s_0) := \inf_{u(\cdot)} J(s_0, u)$.

Assumption 5. The value function $V^* : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ is continuously differentiable on $\mathbb{R}^{n+1}$, i.e., $V^* \in C^1(\mathbb{R}^{n+1}; \mathbb{R})$.

Under Assumption 5, V^* is the unique solution of the stationary Hamilton–Jacobi–Bellman (HJB) equation (Kamalapurkar et al., 2018, Theorem 1.5).

$$\inf_{u \in \mathbb{R}^m} \{ \nabla V^*(s)^{\top} (A(s)\theta + F(s) + G(s)u) + s^{\top} Q s + u^{\top} R u \} = 0, \quad (16)$$

for all $s \in \mathbb{R}^{n+1}$. The Hamiltonian associated with (16) is strictly convex in u due to $R \succ 0$. Thus, for each $s \in \mathbb{R}^{n+1}$, the minimizing control is uniquely determined by the first-order optimality condition $0 = \frac{\partial}{\partial u} (\nabla V^*(s)^{\top} G(s)u + u^{\top} R u) = G(s)^{\top} \nabla V^*(s) + 2Ru$, which yields the optimal feedback control policy $u^*(s) = -\frac{1}{2}R^{-1}G(s)^{\top} \nabla V^*(s)$. Substitution of u^* into (16) yields the HJB equation

$$\begin{aligned} & \nabla V^*(s)^{\top} (F(s) + A(s)\theta) + s^{\top} Q s \\ & - \frac{1}{4} \nabla V^*(s)^{\top} G(s) R^{-1} G(s)^{\top} \nabla V^*(s) = 0. \end{aligned} \quad (17)$$

5.2 Value Function Approximation

For general nonlinear systems, the HJB equation in (14)–(17) does not always have a closed-form solution. Hence,

we use an ADP approximation on a compact set $\Omega \subset \mathbb{R}^{n+1}$, with $\mathcal{C} \subset \Omega$, following Bhasin et al. (2012).

The value function is approximated within a compact set Ω as

$$V^*(s) = W^\top \sigma(s) + \epsilon(s), \quad (18)$$

where $W \in \mathbb{R}^L$ is an unknown vector of bounded weights, $\sigma : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^L$ is a continuously differentiable vector of nonlinear activation functions satisfying $\sigma(0) = 0$ and $\nabla_x \sigma(0) = 0$, $L \in \mathbb{N}$ is the number of basis functions, and $\epsilon : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ denotes the reconstruction error.

By (Sauvigny, 2012, Theorem. 1.5), the parameters L and $\sigma(\cdot)$ can be selected such that the weights and approximation errors satisfy $\sup_{x \in \Omega} \|\epsilon(\cdot)\| \leq \bar{\epsilon}$, $\sup_{x \in \Omega} \|\nabla(\cdot)\epsilon(\cdot)\| \leq \bar{\epsilon}$, $\sup_{x \in \Omega} \|W\| \leq \bar{W}$, $\sup_{x \in \Omega} \|\sigma(\cdot)\| \leq \bar{\sigma}$, $\sup_{x \in \Omega} \|\nabla(\cdot)\sigma(\cdot)\| \leq \bar{\nabla}\sigma$,

Since the weight vector W is unknown, both the value function and the feedback policy are approximated by parametrized functions. Let $\hat{W}_c, \hat{W}_a \in \mathbb{R}^L$ denote adaptive weight estimates. The critic is defined by

$$\hat{V}(s, \hat{W}_c) := \hat{W}_c^\top \sigma(s), \quad \forall s \in \Omega, \quad (19)$$

and the actor is defined by

$$\hat{u}(s, \hat{W}_a) := -\frac{1}{2} R^{-1} G(s)^\top \nabla \sigma(s)^\top \hat{W}_a, \quad \forall s \in \Omega. \quad (20)$$

5.3 Bellman Error

Using the critic $\hat{V}$ and actor $\hat{u}$, the temporal difference, also referred to as the Bellman error (BE), is defined as

$$\begin{aligned} \hat{\delta}(s, \hat{\theta}, \hat{W}_a, \hat{W}_c) &:= s^\top Qs + \hat{u}(s, \hat{W}_a)^\top R \hat{u}(s, \hat{W}_a) \\ &+ \nabla \hat{V}(s, \hat{W}_c)^\top (A(s)\hat{\theta} + F(s) + G(s)\hat{u}(s, \hat{W}_a)). \end{aligned} \quad (21)$$

Using the approximate representation (19)–(20), we introduce the critic weight error and actor weight error as $\tilde{W}_c := W - \hat{W}_c$ and $\tilde{W}_a := W - \hat{W}_a$. For the subsequent analysis, the BE can be expressed in terms of the errors $\tilde{W}_c$ and $\tilde{W}_a$ as

$$\hat{\delta} = -\omega^\top \tilde{W}_c + \frac{1}{4} \tilde{W}_a^\top G_\sigma \tilde{W}_a - W^\top \nabla \sigma A \tilde{\theta} + \Delta, \quad (22)$$

where $\omega := \nabla \sigma (A(s)\hat{\theta} + F(s) + G(s)\hat{u}(s, \hat{W}_a))$, $G_R := GR^{-1}G^\top$, $G_\epsilon := \nabla \epsilon G_R \nabla \epsilon^\top$, $G_\sigma := \nabla \sigma G_R^{-1} G^\top \nabla \sigma^\top$, $\Delta := \frac{1}{2} W^\top \nabla \sigma G_R \nabla \epsilon^\top + \frac{1}{4} G_\epsilon - \nabla \epsilon F$, $A := A(s)$, $F := F(s)$, $\epsilon := \epsilon(s)$, and $\sigma := \sigma(s)$.

Online RL schemes typically require persistence of excitation (PE) for parameter convergence, but PE is difficult to enforce or verify online. When a system model is available, BE extrapolation provides virtual excitation: the BE can be evaluated at selected off-trajectory states $\{s_k(t)\}_{k=1}^N$, and a PE-like condition can be checked via a minimum-eigenvalue criterion. The extrapolated BE is defined as

$$\hat{\delta}_k := \hat{\delta}(s_k, \hat{\theta}, \hat{W}_a, \hat{W}_c), \quad (23)$$

and admits the representation

$$\hat{\delta}_k = -\omega_k^\top \tilde{W}_c + \frac{1}{4} \tilde{W}_a^\top G_{\sigma_k} \tilde{W}_a - W^\top \nabla \sigma_k A_k \tilde{\theta} + \Delta_k, \quad (24)$$

where $\omega_k := \nabla \sigma_k (A_k \hat{\theta} + F_k + G_k \hat{u}(s_k, \hat{W}_a))$, $G_{R_k} := G_k R^{-1} G_k^\top$, $G_{\sigma_k} := \nabla \sigma_k G_R^{-1} G^\top \nabla \sigma_k^\top$, $G_{\epsilon_k} := \nabla \epsilon_k G_{R_k} \nabla \epsilon_k^\top$,

$\Delta_k := \frac{1}{2} W^\top \nabla \sigma_k G_{R_k} \nabla \epsilon_k^\top + \frac{1}{4} G_{\epsilon_k} - \nabla \epsilon_k F_k$, $A_k := A(s_k)$, $F_k := F(s_k)$, $\epsilon_k := \epsilon(s_k)$, and $\sigma_k := \sigma(s_k)$.

Under Assumption 1 and the universal approximation property of neural networks, the Bellman-error residual satisfies $\sup_{s \in \Omega} \|\Delta(s)\| \leq \bar{\Delta}$ for some constant $\bar{\Delta} > 0$. Likewise, for any extrapolated state $s_k \in \Omega$, the corresponding residual obeys $\|\Delta_k\| \leq \bar{\Delta}$.

5.4 Update Laws for Actor and Critic Weights

The critic update law is obtained by applying a gradient-type adaptation to the squared BE. This yields

$$\dot{\hat{W}}_c = -k_{c_1} \Upsilon \frac{\omega}{\rho} \hat{\delta} - \frac{k_{c_2}}{N} \Upsilon \sum_{k=1}^M \frac{\omega_k}{\rho_k} \delta_k, \quad (25)$$

where $k_{c_1}, k_{c_2} > 0$ are adaptation gains and the normalization factors $\rho = 1 + \nu \omega^\top \omega$ and $\rho_k = 1 + \nu \omega_k^\top \omega_k$ use a regularization constant $\nu > 0$. The gain matrix $\Upsilon : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{L \times L}$ follows a standard recursive least-squares update,

$$\dot{\Upsilon} = \beta_c \Upsilon - k_{c_1} \Upsilon \frac{\omega \omega^\top}{\rho^2} \Upsilon - \frac{k_c}{N} \Upsilon \sum_{i=1}^M \frac{\omega_i \omega_i^\top}{\rho_i^2} \Upsilon, \quad (26)$$

with $\Upsilon(0) = \Upsilon_0$ and forgetting factor $\beta_c > 0$. To guarantee parameter convergence, the subsequent stability analysis employs the following excitation condition.

Assumption 6. There exists a constant $\underline{c}_1 > 0$ such that the trajectories $\{s_k(\cdot)\}_{k=1}^N$ satisfy

$$\underline{c}_1 \leq \inf_{t \geq 0} \lambda_{\min} \left(\frac{1}{N} \sum_{k=1}^M \frac{\omega_k(t) \omega_k^\top(t)}{\rho_k^2(t)} \right). \quad (27)$$

As shown in Kamalapurkar et al. (2016a) Lemma 1, provided Assumption 6 holds and $\lambda_{\min}\{\Upsilon_0^{-1}\} > 0$, the update law in (26) ensures that the least squares gain satisfies

$$\underline{\Upsilon} I_L \preceq \Upsilon(t) \preceq \bar{\Upsilon} I_L, \quad (28)$$

$\forall t \in \mathbb{R}_{\geq 0}$ and for some $\bar{\Upsilon}, \underline{\Upsilon} > 0$.

The actor update law is obtained by gradient descent on a regularized loss formed from the BE. The resulting adaptation law is

$$\dot{\hat{W}}_a = -k_{a_1} (\hat{W}_a - \hat{W}_c) - k_{a_2} \hat{W}_a + \frac{k_{c_1} G_\sigma^\top \hat{W}_a \omega^\top}{4\rho} \hat{W}_c + \sum_{k=1}^M \frac{k_c G_{\sigma_k}^\top \hat{W}_a \omega_k^\top}{4N \rho_k} \hat{W}_c,$$

where $k_{a_1}, k_{a_2} > 0$ are adaptation gains. The control input applied to the system (14) uses weights in (25) as

$$u(t) = \hat{u}(s(t), \hat{W}_a(t)), \quad t \geq 0. \quad (29)$$

To satisfy Assumption 4, data may be obtained either from a previous system run or by operating the system under a known stabilizing controller for a finite duration until the recorded data satisfies the required rank condition; see Kamalapurkar et al. (2016b). The adaptive actor-critic update laws presented in this section will be utilized in the subsequent stability analysis to prove the uniform ultimate boundedness of the trajectories of the closed-loop system.

6. STABILITY ANALYSIS

To facilitate the analysis, we define the concatenated vector $Z := [s^\top, \tilde{z}^\top, \tilde{\theta}^\top, \tilde{W}_c^\top, \tilde{W}_a^\top]^\top \in \mathbb{R}^{n+p+2L+2}$ as the

state of the closed-loop system, and define a continuously differentiable candidate Lyapunov function as

$$V_L(Z, t) := V^*(s) + \frac{1}{2}\tilde{z}^\top \tilde{z} + \frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}(t)\tilde{\theta} + \frac{1}{2}\tilde{W}_c^\top \Upsilon^{-1}(t)\tilde{W}_c + \frac{1}{2}\tilde{W}_a^\top \tilde{W}_a \quad (30)$$

From (28) and the positive definiteness of (30), (Khalil, 2002, Lemma 4.3) implies

$$\underline{v}_l(\|Z\|) \leq V_L(Z, t) \leq \bar{v}_l(\|Z\|), \quad (31)$$

for all $t \in \mathbb{R}_{\geq 0}$ and for all $Z \in \mathbb{R}^{n+p+2L+2}$, where $\underline{v}_l, \bar{v}_l : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ are class $\mathcal{K}$ functions. Using the least square gain update law in (26) and the bound in (28), the normalized regressor $\|\frac{\omega(t)}{\rho(t)}\|$ is bounded as $\|\frac{\omega(t)}{\rho(t)}\| \leq \frac{1}{2\sqrt{\nu\Upsilon}}, \forall t \geq 0$. For any continuous function $f : \Omega \rightarrow \mathbb{R}^m$, we define $\bar{f} := \sup_{s \in \Omega} \|f(s)\|$. Using this notation, we define the constants $\varpi_1 := (k_{c1} + k_{c2})\bar{W}\bar{\nabla}\sigma\bar{A}$, $\varpi_2 := (k_{c1} + k_{c2})\bar{\Delta}$, $\varpi_3 := \frac{1}{2}\bar{W}\bar{G}_\sigma + \frac{1}{2}\bar{\nabla}\bar{G}\bar{\nabla}\sigma + k_{a2}\bar{W} + \frac{1}{4}(k_{c1} + k_{c2})\bar{W}^2\bar{G}_\sigma$, $\varpi_4 := k_{a1} + \frac{1}{4}(k_{c1} + k_{c2})\bar{W}\bar{G}_\sigma$, $\varpi_5 := \frac{1}{4}(k_{c1} + k_{c2})\bar{W}\bar{G}_\sigma$. Also, let the positive constant residual be defined as $\iota := \frac{\varpi_2^2}{2\varepsilon_2} + \frac{(\varpi_3 + \varpi_4)^2}{2\varepsilon_3} + \frac{1}{4}\bar{G}_\epsilon$. The following theorem establishes local uniform ultimate boundedness (UUB) of the trajectories of the closed-loop system.

Theorem 1. Let $\bar{\mathbb{B}}_\chi \subset \Omega \times \mathbb{R}^{p+2L+1}$ be a closed ball of radius $\chi > 0$ containing the origin. Provided Assumptions 1–6 hold, if the unknown parameters $\hat{\theta}$ are estimated using the adaptive update law in (11) and (4), the sufficient conditions in (34) and $\underline{v}_l^{-1}(\iota) \leq \bar{v}_l^{-1}(\underline{v}_l(\chi))$ is satisfied, and the weights $\hat{W}_c, \Upsilon, \hat{W}_a, \hat{\theta}$, and Γ are updated according to (25), (26), (5.4), (11), and (4) respectively, then the concatenated state, Z , is locally uniformly ultimately bounded under the controller designed in (29). In particular, with $\mathcal{S}_0$ defined as

$$\mathcal{S}_0 := \{x(0) \in \mathcal{S} \mid Z(0) \leq \bar{v}_l^{-1}(\underline{v}_l(\chi))\}, \quad (32)$$

every trajectory with $x(0) \in \mathcal{S}_0$, satisfies $x(t) \in \mathcal{S}, \forall t \geq 0$.

Proof. The orbital derivative of (30) along the trajectories of the closed loop system is given by $\dot{V}_L(Z, t)$. By substituting (8), (4), (25), (4), and (5.4), then by the property of projection operators (Krstic et al., 1995, Lemma E.1. IV), using Cauchy-Schwarz inequality, completion of squares, and applying Young's inequality, $\dot{V}_L$ can be bounded as

$$\dot{V}_L(Z, t) \leq -\alpha_s\|s\|^2 - \alpha_z\|\tilde{z}\|^2 - \alpha_\theta\|\tilde{\theta}\|^2 - \alpha_c\|\tilde{W}_c\|^2 - \alpha_a\|\tilde{W}_a\|^2 + \iota \quad (33)$$

To make the quadratic part strictly negative definite, choose $\varepsilon_1, \varepsilon_2, \varepsilon_3 > 0$, and gains such that the effective quadratic coefficients

$$\begin{aligned} \alpha_s &:= \lambda_{\min}(Q) > 0, \quad \alpha_z := \gamma > 0, \\ \alpha_\theta &:= (k_\theta - 1)\underline{\sigma}_\theta + \frac{\beta_\theta}{k_\theta\bar{\Gamma}} - \frac{\varpi_1^2}{2\varepsilon_1} > 0, \\ \alpha_c &:= k_{c2}\underline{c} - \frac{\varepsilon_1}{2} - \frac{\varepsilon_2}{2} > 0, \quad \alpha_a := (k_{a1} + k_{a2}) - \varpi_5 - \frac{\varepsilon_3}{2} > 0. \end{aligned} \quad (34)$$

where $\underline{c} := \frac{\beta_c}{2\Upsilon k_{c2}} + \frac{\varepsilon_1}{2}$ is a positive constant and $\varepsilon_1, \varepsilon_2, \varepsilon_3 > 0$ are positive scalars.

If the gain values $k_\theta, k_{c2}, k_{a1}, k_{a2}$ are chosen so that the right-hand sides above are positive, then $\alpha_s, \alpha_z, \alpha_\theta, \alpha_c, \alpha_a$ are all positive. Under these choices we obtain from (33)

$$\dot{V}_L \leq -\underline{v}_l(\|Z\|), \quad \forall \|Z\| \geq \underline{v}_l^{-1}(\iota), \quad (35)$$

for all $t \in \mathbb{R}_{\geq 0}$, and for all $Z \in \bar{\mathbb{B}}_\chi$, where $\underline{v}_l : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a class $\mathcal{K}$ function that satisfies

$$\underline{v}_l(\|Z\|) \leq \frac{1}{2} \min\{\alpha_s, \alpha_z, \alpha_\theta, \alpha_c, \alpha_a\} \|Z\|^2. \quad (36)$$

If the conditions in (34) hold and the bound in (31) is maintained, Theorem 4.18 of Khalil (2002) ensures local UUB of Z . Any solution with $\|Z(0)\| \leq \bar{v}_l^{-1}(\underline{v}_l(\chi))$ obeys $\limsup_{t \rightarrow \infty} \|Z(t)\| \leq \underline{v}_l^{-1}(\bar{v}_l(\underline{v}_l^{-1}(\iota)))$. Moreover, the concatenated state trajectories remain confined to the ball $\bar{\mathbb{B}}_\chi$ for all $t \in \mathbb{R}_{\geq 0}$. Since $\tilde{W}_a$ is UUB, $\hat{u}$ constitutes an approximation of the optimal policy u^* . Hence, $\forall x(0) \in \mathcal{S}_0$, $z(t)$ remains bounded $\forall t \geq 0$. Since boundedness of $z(t)$ is equivalent to $x(t) \in \mathcal{S}$ by Lemma 1, $x(t)$ remains in $\mathcal{S}, \forall t \geq 0$. $\square$

Theorem 2. If Assumptions 1 - 4 hold, if the unknown parameters and the state estimates are updated using (7) and (11), and the gains are selected such that

$$\frac{\beta_\theta}{k_\theta\bar{\Gamma}} > \underline{\sigma}_\theta, \quad (37)$$

then the error systems in (8) and (4) are exponentially stable.

Proof. Let a candidate Lyapunov function be defined as,

$$\tilde{V}(\tilde{z}, \tilde{\theta}, t) = \frac{1}{2}\tilde{z}^\top \tilde{z} + \frac{1}{2}\tilde{\theta}^\top \Gamma(t)^{-1}\tilde{\theta}, \quad (38)$$

then $\exists \underline{\tilde{v}}, \bar{\tilde{v}} > 0$ such that $\underline{\tilde{v}}\|(\tilde{z}, \tilde{\theta})\|^2 \leq \tilde{V}(\tilde{z}, \tilde{\theta}, t) \leq \bar{\tilde{v}}\|(\tilde{z}, \tilde{\theta})\|^2, \forall t \geq 0$, which holds locally by positive definiteness of $\tilde{V}$ and continuity of $\Gamma(t)$. By the property of projection operators in (Krstic et al., 1995, Lemma E.1. IV), the orbital derivative of $\tilde{V}$ along the trajectories (8), (4), (4), and using Assumption 4, applying the triangle inequality, and provided the gain condition in (37) is satisfied, is bounded as

$$\dot{\tilde{V}}(\tilde{z}, \tilde{\theta}, t) \leq -\gamma\|\tilde{z}\|^2 - k_\theta\underline{\sigma}_\theta\|\tilde{\theta}\|^2. \quad (39)$$

Hence, by Theorem 4.8 of Khalil (2002), the error dynamics (8) and (4) are exponentially stable. Consequently, under the ICL-based update law (11), the parameter estimates $\hat{\theta}(t)$ converge exponentially to the true parameters θ , i.e., $\lim_{t \rightarrow \infty} \tilde{\theta}(t) = \lim_{t \rightarrow \infty} (\hat{\theta}(t) - \theta) = 0$. $\square$

7. SIMULATION STUDY

This section evaluates the developed policy on an obstacle-avoidance problem with dynamics of the form (1), state $x = [x_1, x_2]^\top$, and $Y(x) = [x_1, x_2, 0, 0; 0, 0, x_1 + x_2, x_1^2 x_2]$, where $g(x) = [0, \cos(2x_1) + 2]^\top$, $f(x) = 0$, and $\theta = [-1, -1, -0.5, -0.5]^\top$. The safe set is defined by $h(x) = (x_1 - 1)^2 + (x_2 - 2)^2 - 0.5^2$, with $Q = I_3$ and $R = 1$ in (15). The initial conditions are $x(0) = [2.5, 4]^\top, \hat{z}(0) = 0, \hat{\theta}(0) = 0_{4 \times 1}, \hat{W}_c(0) = \hat{W}_a(0) = 0.5\mathbf{1}_{6 \times 1}, \Gamma(0) = 10I_4$, and $\Upsilon(0) = 0.01I_6$. The barrier function is chosen as $\beta(x) = B(h(x)) = K/h(x)$, so that $\Phi(\beta) = -K/\beta^2$, with $K = 0.01$ and observer gain $\gamma = 3$. The ADP gains are $\nu = 2, k_{c1} = k_{c2} = 1, k_{a1} = 2, k_{a2} = 1$, and $\beta_c = 0.1$, with $\Upsilon = 0$ and $\bar{\Upsilon} = 1000$. The ICL gains are $k_\theta = 50, \kappa = 1$, and $\beta_\theta = 1$. The value-function basis is $\sigma(s) = [s_1^2 \ s_2^2 \ s_3^2 \ s_1 s_2 \ s_2 s_3 \ s_3 s_1]^\top$ where $s = [x_1, x_2, z]^\top$. The simulation uses 100 fixed Bellman-error extrapolation points s_k distributed in a 4×4 square centered at the

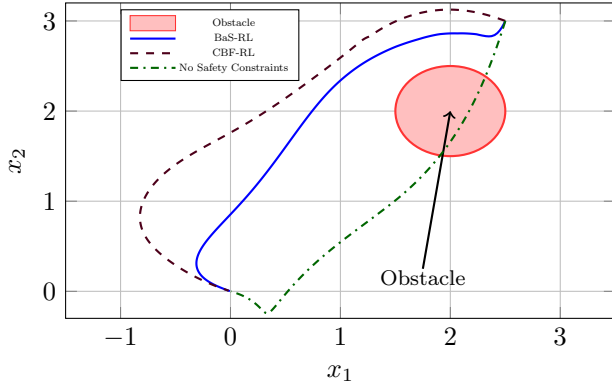


Fig. 1. Phase-space trajectories of the system

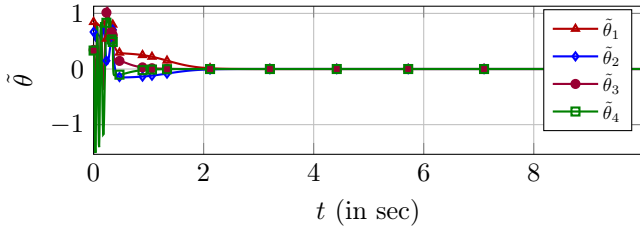


Fig. 2. Evolution of the $\tilde{\theta}$ with time.

origin. We compare the proposed BaS-RL-ICL framework with the CBF-RL-ICL method of Cohen et al. (2023) and RL-ICL without safety constraints.

7.1 Discussion

Figure 1 shows the ability of the BaS-RL-ICL policy to successfully avoid the obstacle. Figure 2 shows exponential convergence of the parameter error to zero (Theorem 2). These results highlight the effectiveness of the developed BaS-ICL-RL framework to estimate the parameters while simultaneously ensuring safety.

8. CONCLUSION

In this paper, we developed a BaS-based safe control framework for nonlinear control-affine systems with unknown parameters. Results show the developed policy ensures safety without prior parameter knowledge, while simultaneously driving parameter estimates to their true values, and keeping the system stable. However, this policy is only a near optimal solution, requires finite excitation, and increases nonlinearity and computation cost with the addition of additional BaS states. Future work will address more complex settings like time varying obstacles, dynamic disturbances and global safety guarantees under weaker requirements.

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